CS 6210 Fall 2026 Test 2

(120 min Canvas Quiz)

[Total Points: 78]

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Distributed Systems [24 points]

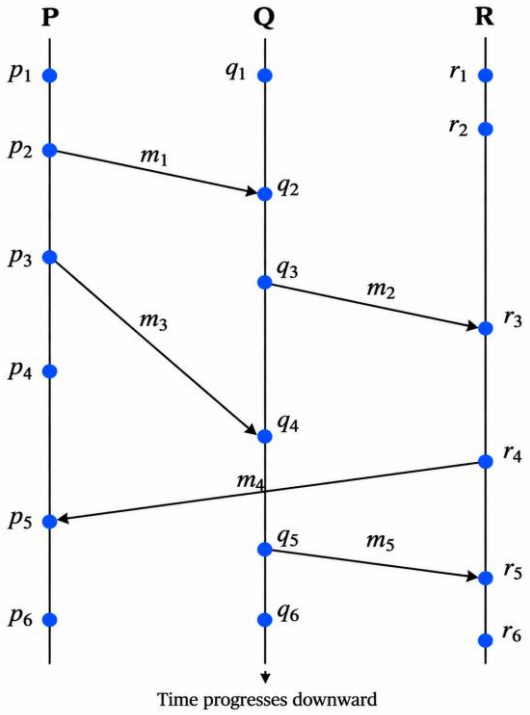
Lamport's Logical Clock [10 points]

1. [8 points] Consider a distributed system consisting of three processes P, Q, and R, each maintaining a Lamport logical clock C_p, C_q, and C_r, respectively. All logical clocks are initialized to 0:

$$C_p(p_1) = C_q(q_1) = Cr(r_1) = 0$$

When a clock is incremented, it is incremented by the smallest positive integer necessary to satisfy Lamport's logical clock conditions. Each message carries the sender's logical timestamp.

The following diagram illustrates the execution of events and message exchanges between the three processes.



Determine the following logical clock values. Explain your reasoning for each answer. (No credit without reasoning.)

a) [2 points] What is the value of C_p at event p_3?

b) [2 points] What is the value of C_q at event q_3? What is the relative ordering of the events p_3 and q_3? Explain.

c) [2 points] What is the value of C_r at event r_3?

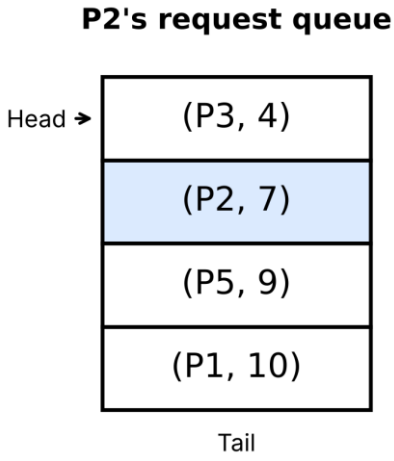
d) [2 points] What is the value of C_p at event p_5?

2. [2 points] Your friend argues that a state-of-the-art multicore CPU is a distributed system per Lamport's definition. Is she right? Justify your answer succinctly. No credit without justification.

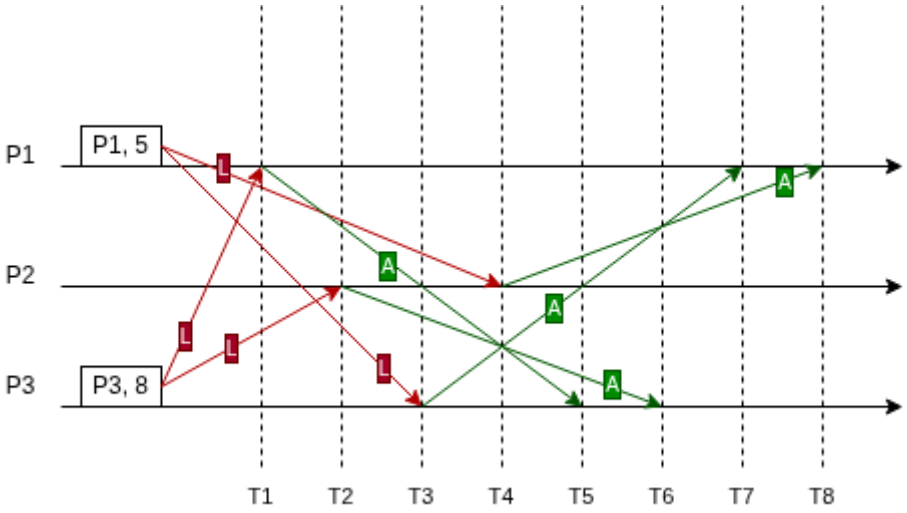
Lamport's ME Algorithm [8 points]

1. [3 points] Consider a distributed system with 5 nodes using Lamport's distributed mutual exclusion algorithm. P2 wishes to acquire a lock. The current state of P2's queue is shown below, and P2 has not received any additional messages. Can P2 acquire the lock? If yes, explain why. Otherwise, what additional messages does P2 need to wait for?

P2's state (top to bottom): (P3, 4), (P2, 7), (P5, 9), (P1, 10)



2. [5 points]



Consider the diagram above. It shows Lamport's mutual lock exclusion lock algorithm in action.

- L is a mutual exclusion lock governing access to a resource.
- P1, P2, and P3 are processes attempting to acquire the lock.
- Time flows in monotonically increasing order horizontally left to right in the figure, where T1 < T2 < … < T7.
- The boxes indicate a request for the lock made by a process. For example, "P1, 5" represents a request made by P1 at timestamp 5.
- Red arrows (L) indicate lock request messages sent to peers
- Green arrows (A) indicate acknowledgement messages from peers for the lock requests
- Assume that no other requests are made other than those shown in the figure.

a) [2 points] With reference to the above timeline of actions, which process will acquire the lock first? And at what timestamp?

b) [3 points] In the figure above, assume that the lock request message from P1 to P3 is lost due to a network glitch. With concise bullets summarize what could lead to both P1 and P3 acquiring the lock simultaneously.

Active Networks [6 points]

1. [3 points] In the ANTS framework, an active router receives a capsule containing a "type" field that it has not seen before.

a) [1 point] How does the active router obtain the code that corresponds to the type field?

b) [2 points] Is it possible that the active router is unable to process the received capsule? How does ANTS deal with capsules that cannot be processed?

2. [3 points]

a) [2 points] Give ONE pro and one CON of ANTS decision NOT to send code in every capsule

b) [1 point] Why is a sandboxed execution environment needed in ANTS framework?

Distributed Objects and Middleware [14 Points]

Spring OS [8 points]

1. [4 points] During a design discussion about remote procedure calls in Spring OS, a colleague argues:

"Building distributed applications on Spring OS is completely effortless because the IDL compiler automatically generates Network Proxies directly from our interface definitions, eliminating the need for any runtime network infrastructure or manual networking code when making cross-node calls."

a) [2 points] Is your colleague's argument valid? If yes, why? If not, what artifacts does IDL compiler provide for the app developer?

b) [2 points] Another colleague argues by saying: *"I disagree. Application developers need to write client-side and server-side proxies on top of application logic to bridge the network, which makes building software for Spring OS difficult."* Do you agree or disagree with your colleague? Justify.

2. [4 points] Alice is reviewing the microkernel communication layer for a high-throughput microservice architecture based on Spring OS. After review she makes the following claim during a design review:

"*Because Spring uses a microkernel, every Door invocation across domains requires the Nucleus to inspect the caller's process ID*

(PID) and ACL permissions to verify whether the client is authorized to make the call. Furthermore, transferring execution between client and server domains incurs a high explicit cost."

a) [2 points] (True/False with Justification. No credit without justification) Is Alice's claim regarding Nucleus PID/ACL inspection on every door call True or False?

b) [2 points] (True/False with Justification. No credit without justification) Is Alice's claim regarding high explicit cost for cross-domain calls correct?

EJB [4 points]

1. An EJB flight-search service initially uses a coarse-grained session bean to perform the business logic for each client request. Within each request, airline lookups run sequentially. A developer replaces this with a finer-grained design in which the servlet coordinates entity-bean calls and runs independent airline lookups concurrently. With the same server hardware and the same airline queries, measurements show lower overall throughput (completed client requests per second).

a) [2 points] Explain how the finer-grained EJB interactions could account for the lower throughput, despite the additional parallelism.

b) [2 points] In the original design, does sequential processing within one client request prevent concurrent processing of requests from different clients? Explain.

Java RMI [2 points]

1. [2 points] A high-availability Java RMI server process is deployed in a shared cloud infrastructure. To support dynamic extensibility, the server dynamically downloads and executes unverified remote stub and parameter classes over the network at runtime. A malicious client attempts a denial-of-service attack by flooding the server with downloaded classes designed to aggressively consume excessive host CPU cycles and memory.

(Answer True or False with justification: No credit without justification) The in-built resource consumption regulation

mechanisms of Java sandboxing will prevent these dynamically loaded classes from abusing host resources.

Distributed Subsystems [26 points]

Global Memory Systems [8 points]

1. [5 points] Consider a GMS cluster with nodes A, B, and C, where the page size is 4KB. A process (P1) running on Node A faults on page Y. Page Y is present in the local memory of Node B, where it is actively being used by a process on Node B. The globally least recently used (LRU) page is currently in the global memory of Node C.

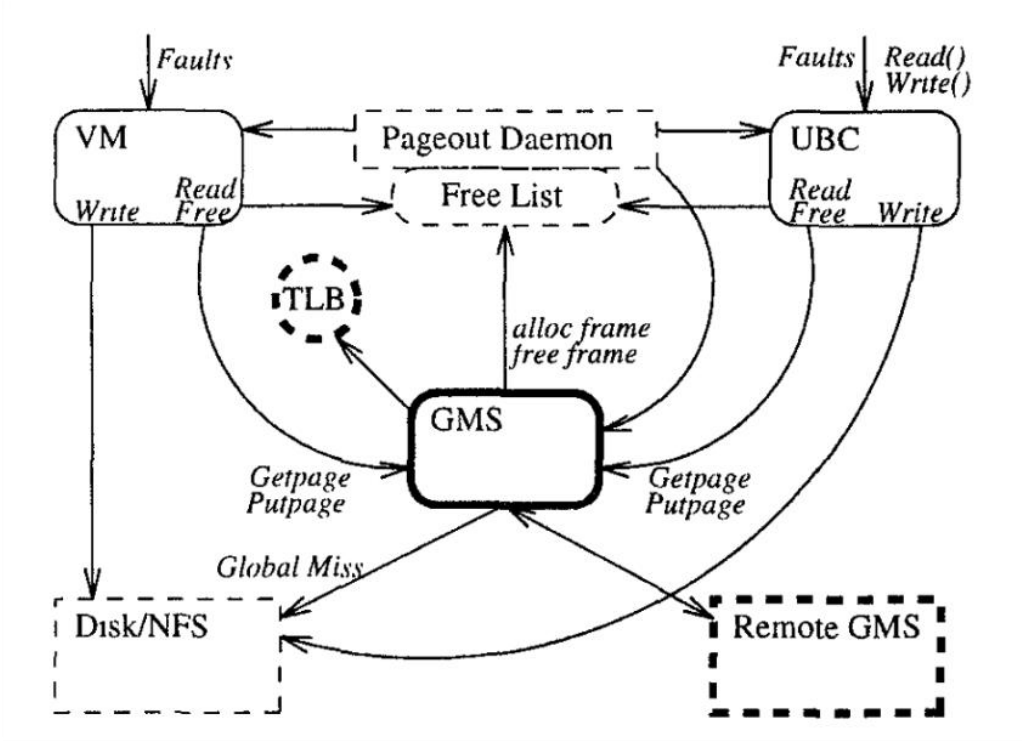
Assume:

- On all nodes, local memory > 0 and global memory > 0.
- GMS uses a strict LRU replacement policy.

a) [3 points] Describe the sequence of steps involved in handling the page fault.

b) [2 points] Quantify the change in local and global memory on Nodes A, B, and C after the fault is handled.

2. [3 points] Refer to the architecture of the modified OSF/1 memory management system shown in the figure



Consider a GMS cluster with nodes A and B.

a) [2 points] A UBC page and a VM page on the same node were last accessed at the same time. For which of the two can GMS easily obtain accurate age information and why? (no points without reasoning)

b) [1 point] Node B crashes while holding pages from Node A in its global memory. Explain why this does not compromise the GMS integrity.

DSM [14 points]

1. [9 points] Consider the following sequence of actions in a TreadMarks DSM system. Assume pristine copies of pages X and Y are at their owner node N0 throughout, and X and Y are initially not present on any other node. T1-T4 are in increasing time order.

T1: N1: acq(L1) write X rel(L1)

T2: N2: acq(L1) read X write Y rel(L1)

T3: N3: acq(L1) write X rel(L1)

T4: N4: acq(L1) read X rel(L1)

a) [2 points] What happens at N2 when it acquires L1 and then reads X?

b) [2 points] What happens when N2 writes Y, and when it releases L1?

c) [2 points] When N4 reads X, whose diffs are applied to X, and in what order?

d) [1 point] Will pristine copy and associated diff for Page Y be brought to N4 during its critical section execution? Explain why or why not.

e) [2 points] Assume each diff is fetched from the node that created it. Counting each request and each reply as distinct messages, how many messages does N4's read of X generate?

2. [5 points] Pages X and Y are shared (other nodes also cache them), but neither is present at N1 at the start. N1 runs the two critical sections below, and no other node acquires L1 in between.

```
acq(L1)
read X
write X
write X
read Y
rel(L1)

acq(L1)
write X
rel(L1)
```

a) [2 points] In the first critical section, how many page faults occur, and how many twins are created?

b) [1 point] How many diffs are created at the first release?

c) [2 points] Does the write to X in the second critical section cause a page fault? Explain.

Distributed File Systems [4 points]

1. [4 points] In xFS, the metadata manager for a file is statically assigned at file creation time. Your teammate Joe proposes the following policy: when a Node N creates a file, assign N as the metadata manager for that file. Another teammate Dick proposes the following policy: choose the metadata manager to be different from the Node N where the file is created.

a) [2 points] Client C created file F and now reads block b of F. The block is not in C's cache, but it is cached at client D. Count each message between two different nodes as one network hop.

(i) How many network hops does it take for b to reach C with Joe's policy?
(ii) How many network hops does it take for b to reach C with Dick's policy?

b) [2 points] Describe one concrete workload where Joe's policy would hurt performance and explain why.

HIDDEN - Potpourri [14 points]